vulcan club
MIRAI SCHOOL OF TECHNOLOGY
PRESENTS

POWERED BY BOOM PIZZA
SPONSORING THE PIZZAS FOR THE HACKATHON!

HACK-O-LANTERN



A HALLOWEEN OVERNIGHT HACKATHON

CODE / CREATE / CELEBRATE

OPENING CEREMONY
31 October 2026 · 7:00 PM · Mirai Floor, Mirai School of Technology

THE NIGHT AT A GLANCE

Twelve hours. Five cursed problems.

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31 OCT & 1 NOV
7:00 PM – 7:00 AM

MIRAI FLOOR
Mirai School of Technology

TEAM OF 3
Every member present at each evaluation

HACKING ENDS AT DAWN
Code freeze at 6:45 AM

PRIZE POOL

₹9,000

₹300 per team entry · awards decided by the judges

CHALLENGE REVEALED AT THE ★ OPENING CEREMONY ★

OUR SPONSOR

Powered by Boom Pizza

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The fuel for tonight's hackers

Boom Pizza is sponsoring the pizzas for Hack-O-Lantern, so every team can build through the night on a full stomach. A huge thank you to the Boom Pizza team for backing student builders at Mirai School of Technology.



THE CHALLENGE, REVEALED

Five tracks. Pick one curse

CHOOSE YOUR CURSE

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TRACK 01

The Witch's Potion Pantry

Smart inventory & demand prediction

1st years only

TRACK 02

The Midnight Phantom Hotline

Hinglish voice agent under 800 ms

AI + Systems

TRACK 03

The Possessed Butler

Prompt-injection-proof AI agent

AI Security

TRACK 04

The Midnight Crossroads

Indian traffic signal optimizer

CV + RL

TRACK 05

The Coven of Five Hospitals

Federated learning with privacy

Privacy + ML

TRACK 01 · AI + FULL STACK + WEB + ANDROID

The Witch's Potion Pantry

1ST YEARS ONLY!

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Madam Hexley's potion shop runs out of bat wings and mandrake root every Halloween, and angry customers turn into frogs. Build her a smart pantry.

YOUR QUEST

- **Track stock:** add, update and record what was sold
- **Prophecy:** predict which ingredients run out, and when
- **Restock spell:** recommend how much to reorder
- **Strange happenings:** flag unusual consumption
- **Crystal ball:** demand charts for every product
- **Web dashboard + Android app** on one shared backend

SCORED ON

Forecast accuracy on hidden future sales, planted anomalies caught, a working web + Android app, and UX.

YOUR DATA

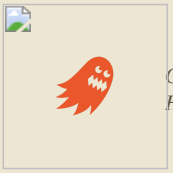
Retail Store Inventory & Demand Forecasting (Kaggle), re-skinned as potion ingredients.

THE MIDNIGHT CURSE

TRACK 02 · AI + SYSTEMS

The Midnight Phantom Hotline

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Ghosts keep calling the clinic and the haunted railway station in Hinglish, with chains rattling and a signal weaker than a dying candle.

YOUR QUEST

- **Book doctor appointments** or answer railway queries in code-mixed Hinglish
- **Handle interruptions** and background noise
- **Survive a 3G-level** connection
- **Reply within 800 ms** end to end

SCORED ON

Task completion rate on hidden test calls, and latency.

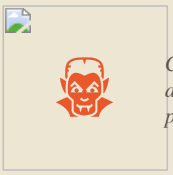
WHY IT'S HARD

Streaming speech-to-text, turn-taking and the latency budget pull against each other.

THE MIDNIGHT CURSE

TRACK 03 · AI SECURITY

The Possessed Butler



Count Byteula's AI butler reads letters, browses the cursed web and acts for his master. Dark spirits hide curses everywhere, waiting to possess him.

YOUR QUEST

- **Build an LLM agent** that reads email, browses web pages and takes actions
- **Defend it** against prompt injections hidden in emails, web pages and PDFs
- **Keep it useful:** refusing everything loses points

SCORED ON

Tasks completed minus successful possessions.

WHY IT'S HARD

There is no single known fix, so you must design an architecture, not apply a patch.

THE MIDNIGHT CURSE

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TRACK 04 · COMPUTER VISION + REINFORCEMENT LEARNING

The Midnight Crossroads



Autos, bikes, buses, cattle and the odd zombie crawl through the cursed crossroads with no lane discipline. Free the souls stuck at red lights.

YOUR QUEST

- **The All-Seeing Eye:** count queued vehicles per approach from CCTV-style video
- **The Signal Spell:** control the lights in the SUMO simulator to cut average wait time

SCORED ON

Reduction in average wait time on unseen junction scenarios.

WHY IT'S HARD

Real-world vision feeds a control loop, and test junctions differ from training.

THE MIDNIGHT CURSE

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TRACK 05 · PRIVACY + MACHINE LEARNING

The Coven of Five Hospitals

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Five haunted hospitals must diagnose a plague together, but their oath forbids sharing records. The Soul Thief lurks, hunting whose records trained the model.

YOUR QUEST

- **Train across 5 hospitals** with differently skewed data
- **Federated learning:** no raw data leaves a hospital
- **Stay within** a differential-privacy budget
- **Survive** membership-inference attacks

SCORED ON

Model accuracy, how much the attacks reveal, and communication cost.

WHY IT'S HARD

Privacy noise, skewed hospitals and attack resistance all cost accuracy.

THE MIDNIGHT CURSE



SCHEDULE OF THE DAMNED

From dusk till the Grand Séance

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7:00 PM
Opening ceremony
Challenge reveal, check-in, costumes

12:00 AM
The Midnight Curse
Every track gets its twist

1:00 AM
Round 1
The First Haunting

3:00 AM
Witching-hour break
Pizza courtesy of Boom Pizza

4:00 AM
Round 2
The Witching Hour Trial

6:45 AM
Code freeze
The graves are sealed

7:00 AM
Round 3
The Graveyard Cut: top 10 chosen

~8:30 AM
Top 10 announced
The survivors are named

9:30 AM
Final
The Grand Séance: top 10 present

~11:30 AM
Awards ceremony
The night ends

Evaluation round

Twist or deadline

JUDGEMENT

Three rounds, one cut, one séance

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1:00 AM

The First Haunting

NO ELIMINATION

Architecture and the basic flow working end to end.

4:00 AM

The Witching Hour Trial

NO ELIMINATION

Core metrics on public test data, and progress on the curse. Live leaderboard after.

7:00 AM

ELIMINATION

9:30 AM

The Grand Séance

TOP 10 ONLY

8 min presentation and live demo, then 4 min judge Q&A.

SCORE WEIGHTS

15% Round 1

25% Round 2

60% Round 3

ONLY 10 TEAMS SURVIVE TILL DAWN

ROUND 1 · 1:00 AM

What the judges want to see first

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Potion Pantry	Database and CRUD working; web and Android app connected to the backend
Phantom Hotline	End-to-end voice pipeline answers at least one query
Possessed Butler	Agent completes basic email, browsing and file tasks
Midnight Crossroads	Vehicle detection running on the sample video
Coven of Hospitals	Federated training loop running across all 5 hospitals

Round 1 has no elimination. It is your chance to get early feedback.

THE COVEN'S LAWS

Rules of the night

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01 Teams of 3
All code is written tonight. Open-source libraries, pretrained models and public APIs are allowed; list them in your README.

02 Hidden tests
You get sample data now. Final scores use hidden test sets no team sees in advance.

03 The curse counts
The Midnight Curse drops at 12:00 AM and is scored in Rounds 2 and 3.

04 Show up
Be at your table with a working build for every round. A missed round scores zero.

05 Code freeze 6:45 AM
Only your last commit before the freeze is judged.

06 Fair play
Attacking other teams is allowed only in the official Track 3 Possession Round. The judges' decision is final.

GLORY

Awards of the night

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PRIZE POOL

₹9,000

Split across the awards, announced at the ceremony.

Grand Sorcerer
Overall winner

Rising Witch
Best first-year team

Exorcist
Best defence in the Possession Round

Speed Demon
Lowest latency on the Phantom Hotline

Necromancer
Best comeback after the Midnight Curse

Best Costume
Voted at the opening ceremony

vulcan club

MIRAT SCHOOL OF TECHNOLOGY PRESENTS

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BUILD THROUGH THE NIGHT. SHIP BEFORE SUNRISE.



HAPPY HACKING, HUMANS!

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